Basic Electrocardiography Course Session 5

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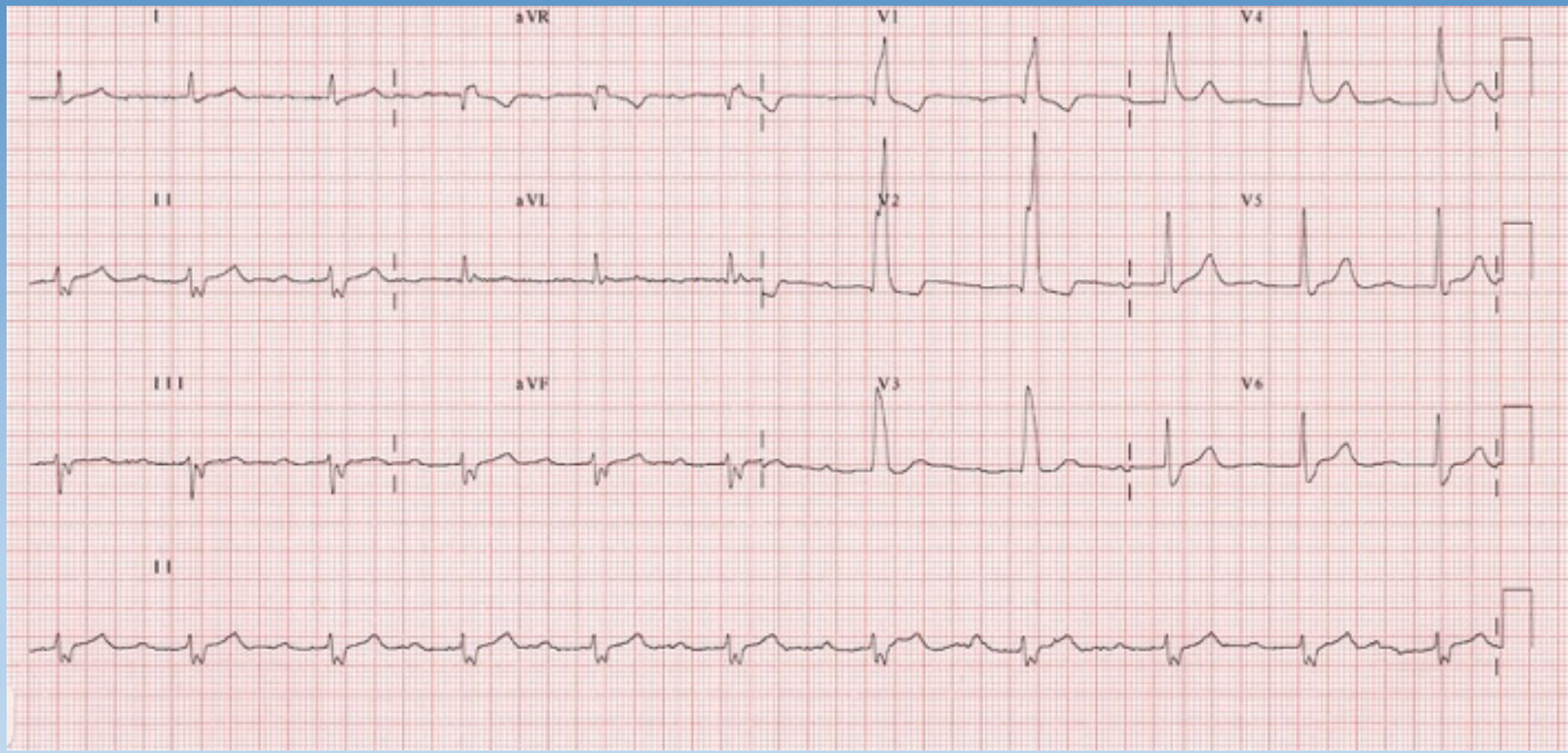
Criteria for Trifascicular Block

- Bifascicular block (RBBB+ LAFB or RBBB+LPFB) plus 1st degree AV block

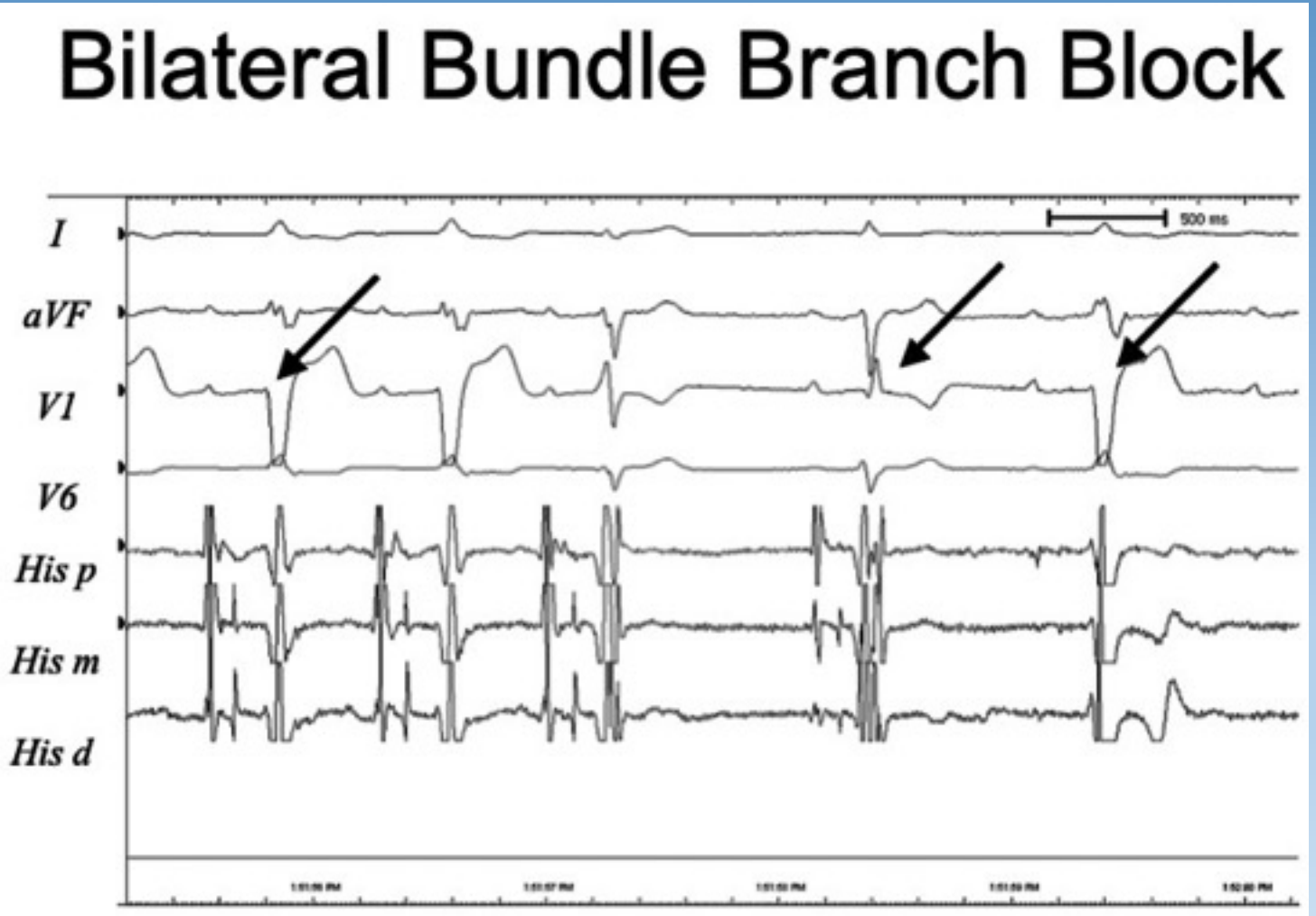
OR

- Alternating bundle branch block

Trifascicular Block



Alternating Bundle Branch Block



Chamber Enlargement

- Left Ventricular Hypertrophy
- Right Ventricular Hypertrophy
- Atrial Enlargement

Left Ventricular Hypertrophy

Voltage Criteria Sufficient for Diagnosis
Without Repolarization Abnormalities

Cornell Criteria:

R wave in aVL + S wave in V3 :

- 28 mm in males
- 20 mm in females

Left Ventricular Hypertrophy

Voltage Criteria Sufficient for Diagnosis Without Repolarization Abnormalities

Other commonly used voltage criteria:

PRECORDIAL LEADS

R wave in V5 or V6 + S wave in V1

>35 mm age > 40 years

>40 mm age 30-40 years

>60 mm age 16-30 years

Maximum R wave + S wave in precordial leads:

>45 mm

Left Ventricular Hypertrophy

Voltage Criteria Sufficient for Diagnosis Without Repolarization Abnormalities

Other commonly used voltage criteria:

LIMB LEADS (one or more)

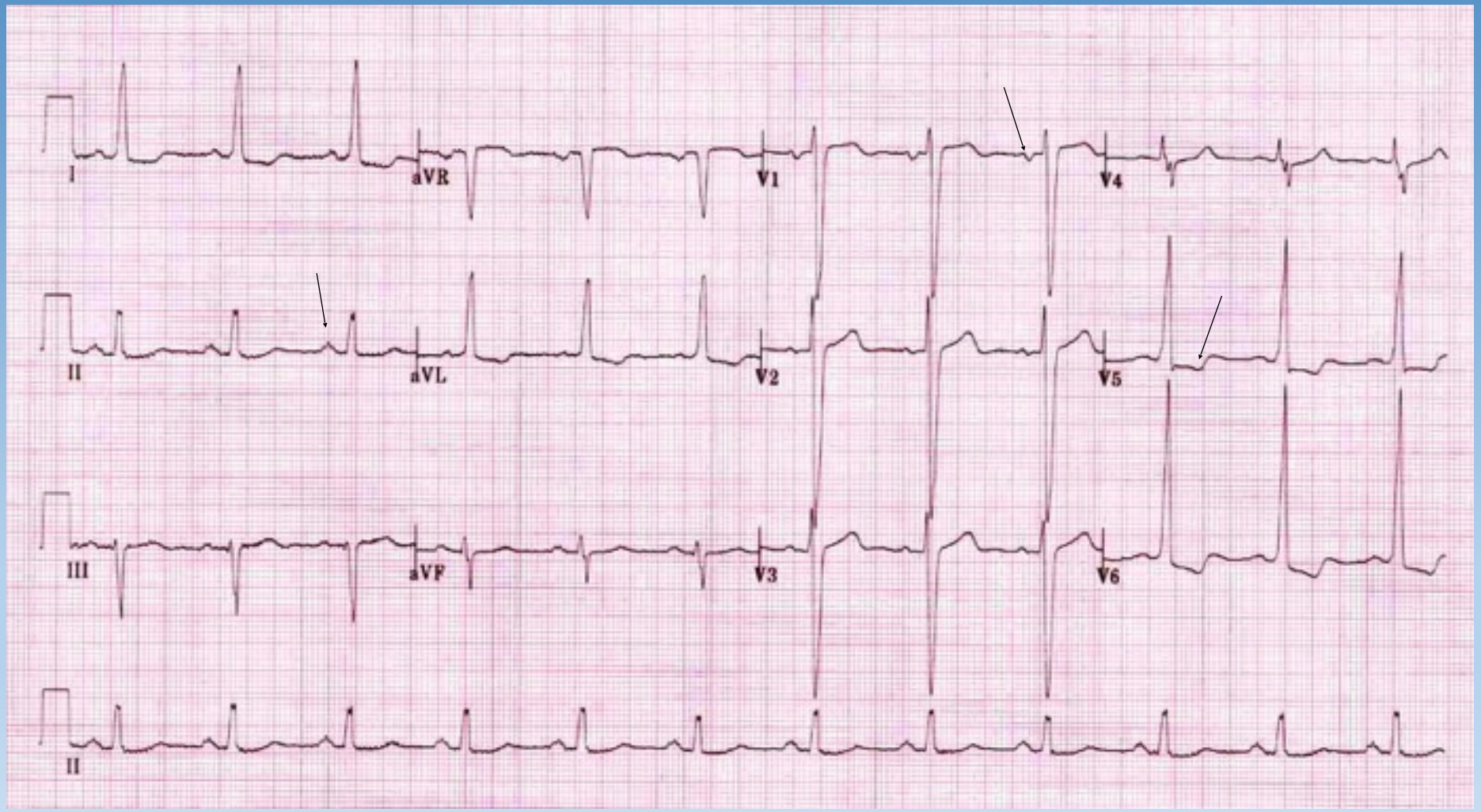
- R wave lead I + S wave lead II > 26mm
- R wave lead I > 14 mm
- S wave aVR > 15 mm
- R wave in aVL > 12 mm (except when LAFB)
- R wave in aVF > 21 mm

Left Ventricular Hypertrophy

Non-voltage related changes (often present but not required for diagnosis of LVH)

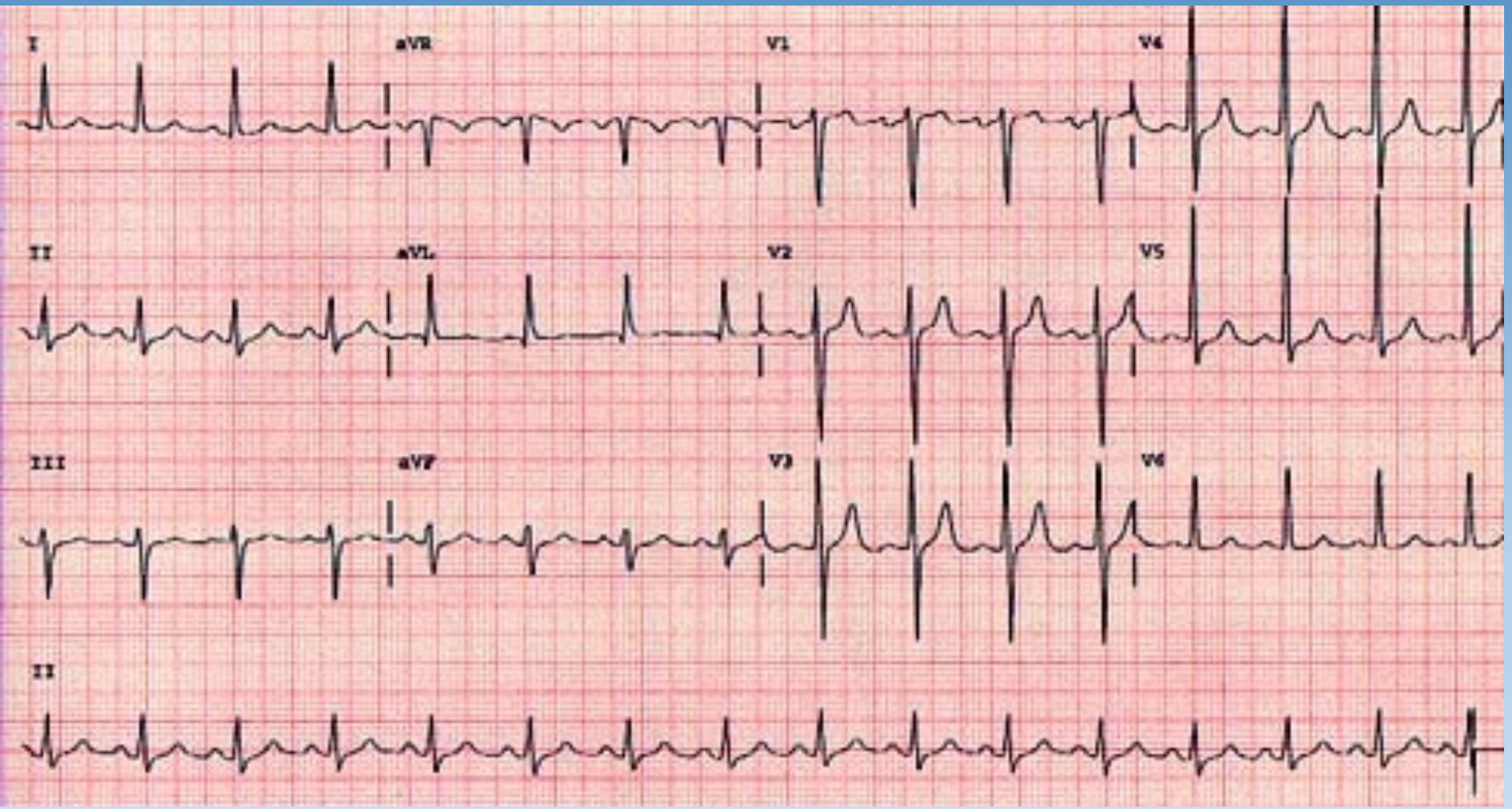
- LAE
- Left axis deviation
- Delayed onset intrinsicoid deflection
- Nonspecific intraventricular conduction disturbance
- Prominent u waves

LVH by Cornell Voltage Criteria
Note also: repolarization abnormalities and LAE



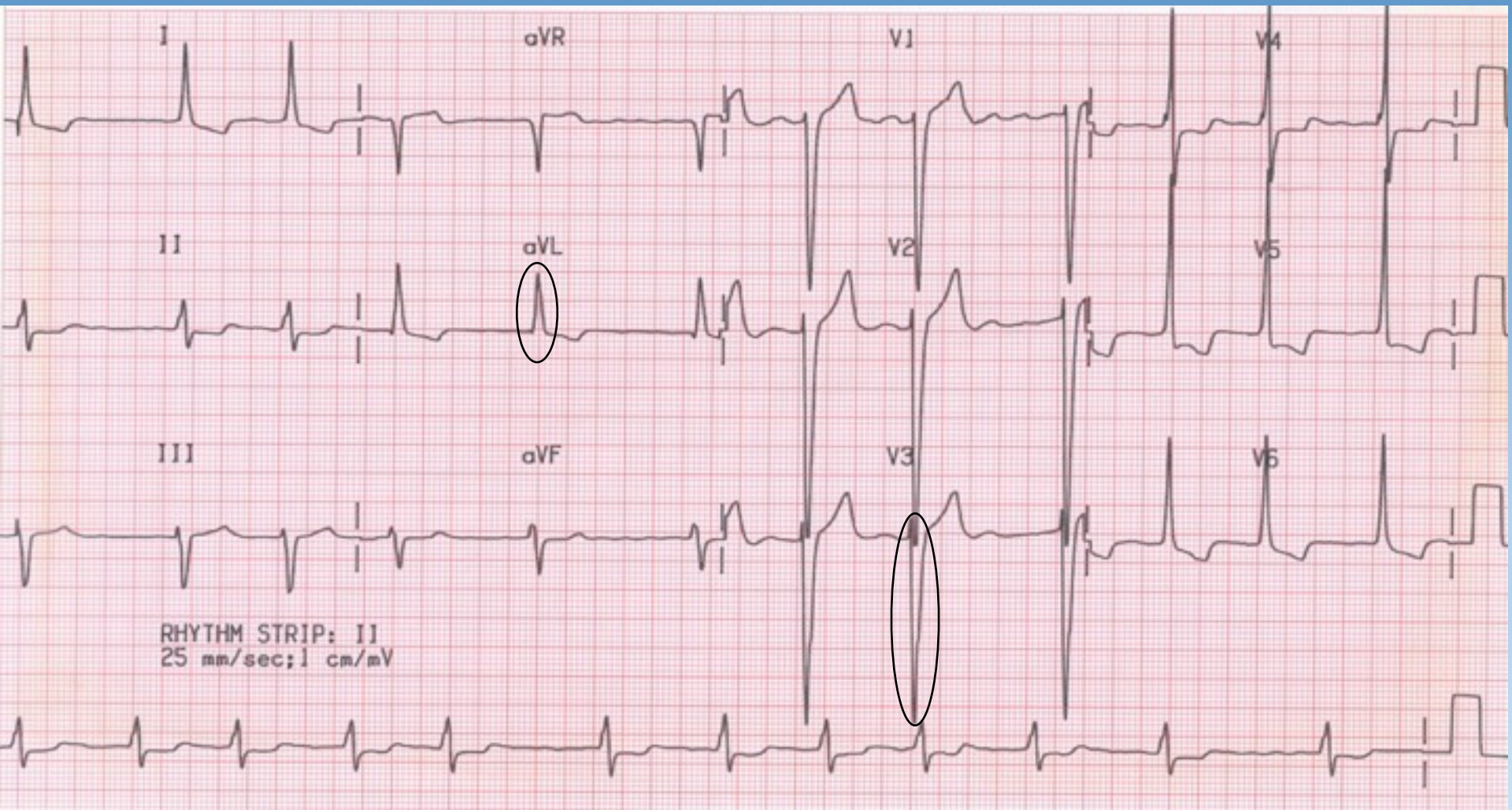
50 year old female

LVH by Cornell Voltage Criteria
Note also LAE and left axis deviation



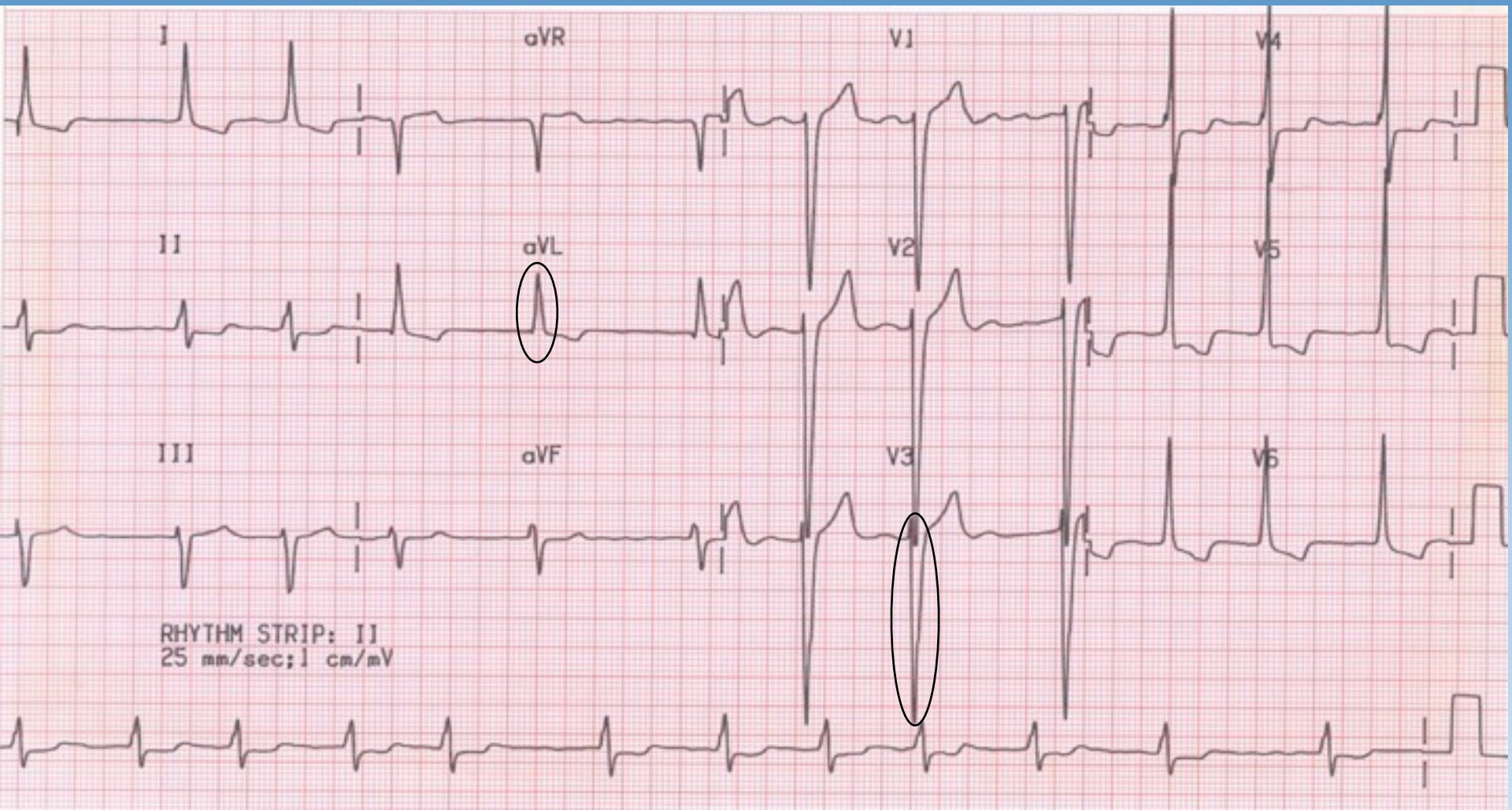
56 yo male

- Rate?
- Rhythm?
- Axis?
- Diagnosis?



56 yo male

- Rate? 60
- Rhythm? Atrial fibrillation
- Axis? Left axis deviation
- Diagnosis? LVH

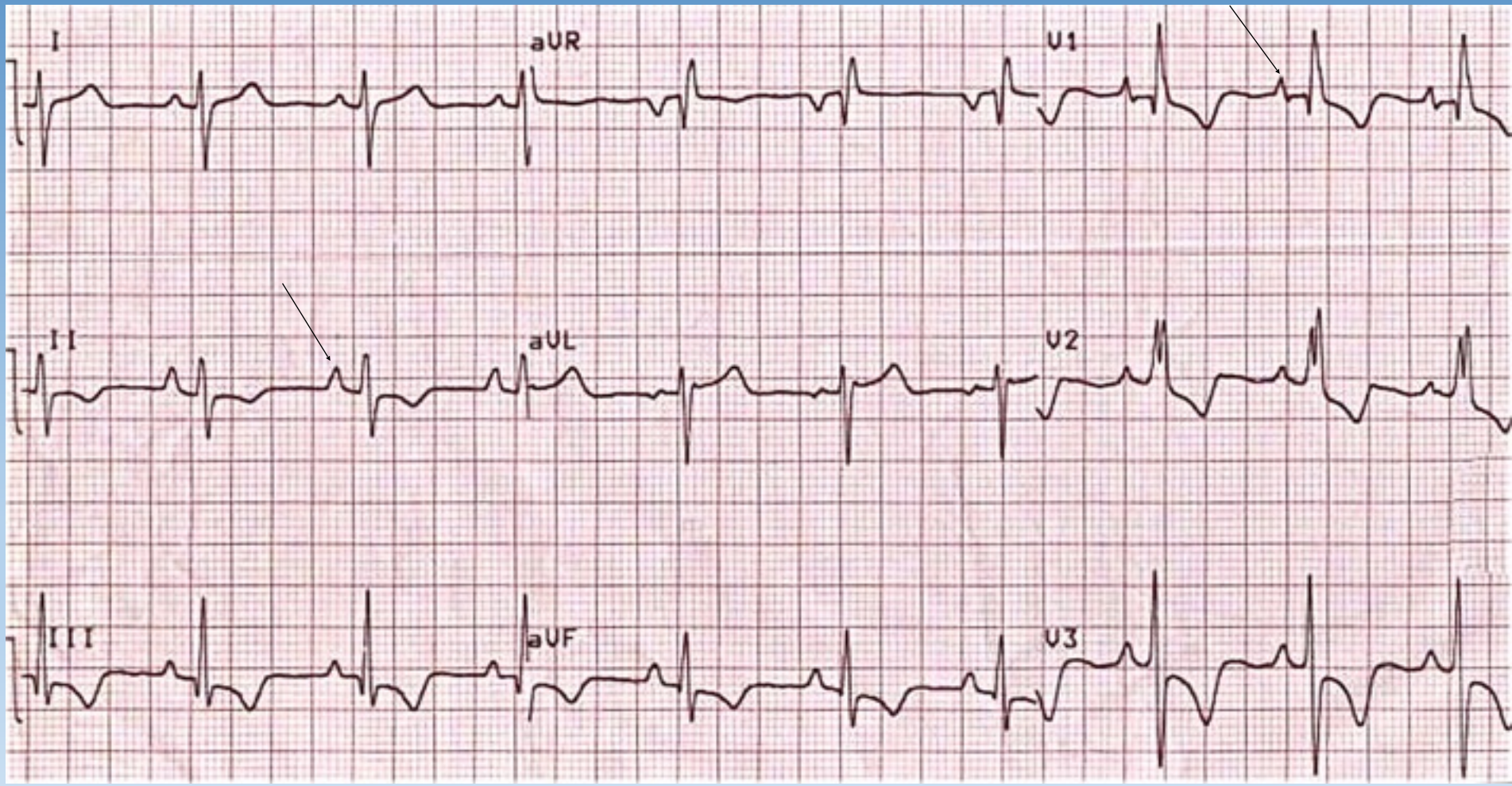


Right Ventricular Hypertrophy

- Right axis deviation
- Dominant R wave
  - ➢ R/S ratio in V1 > 1 or R/S ratio in V5 or V6 < 1
  - ➢ R wave in V1 > 7 mm
  - ➢ R wave in V1 + S wave in V6 > 11
  - ➢ rSR’ in V1 with R’ > 10 mm
  - ➢ qR complex in V1
- Secondary ST-T changes in right precordial leads
- Right atrial abnormality

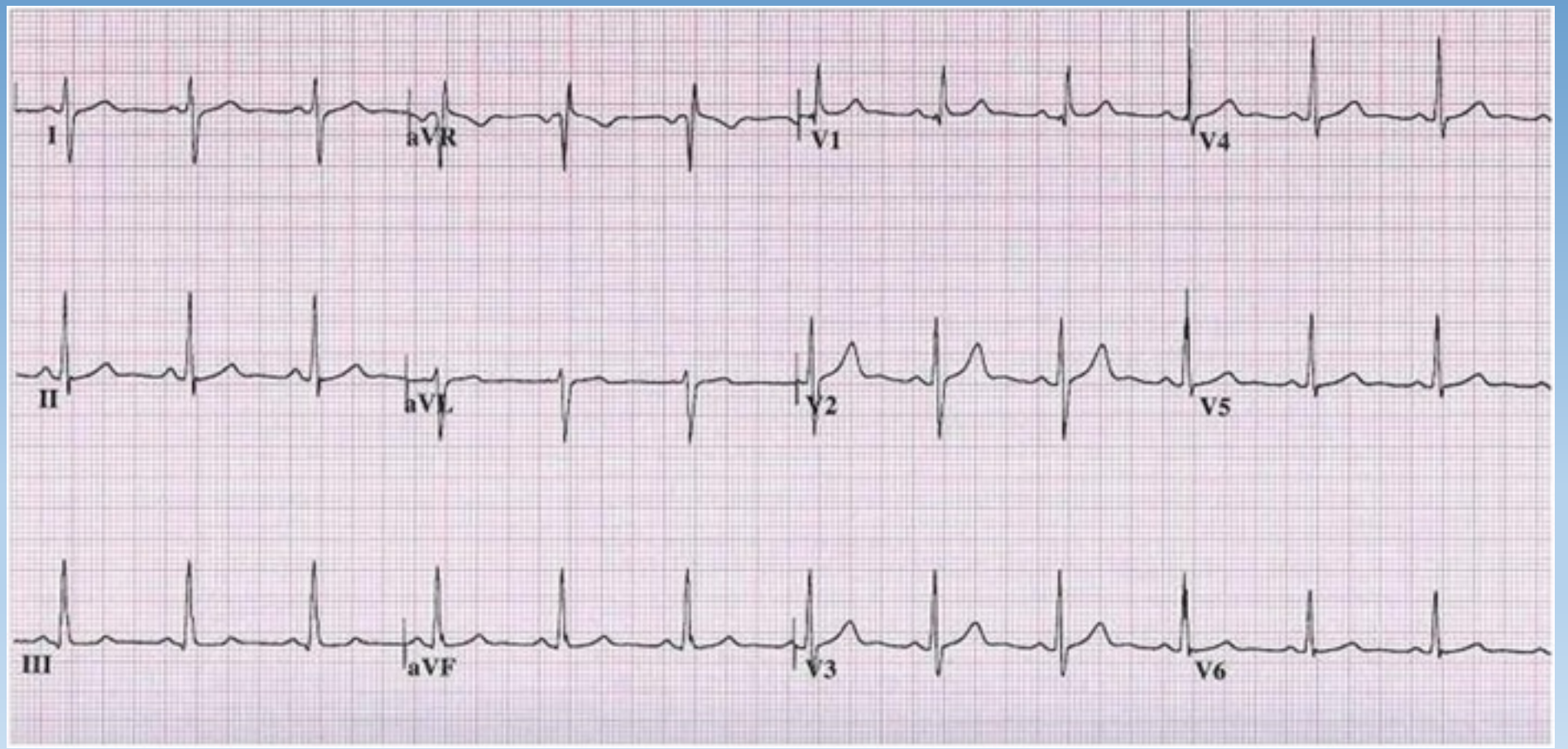
Right Ventricular Hypertrophy

- RAD
- qR complex in V1
- Secondary ST –T changes right precordial leads
- RAE

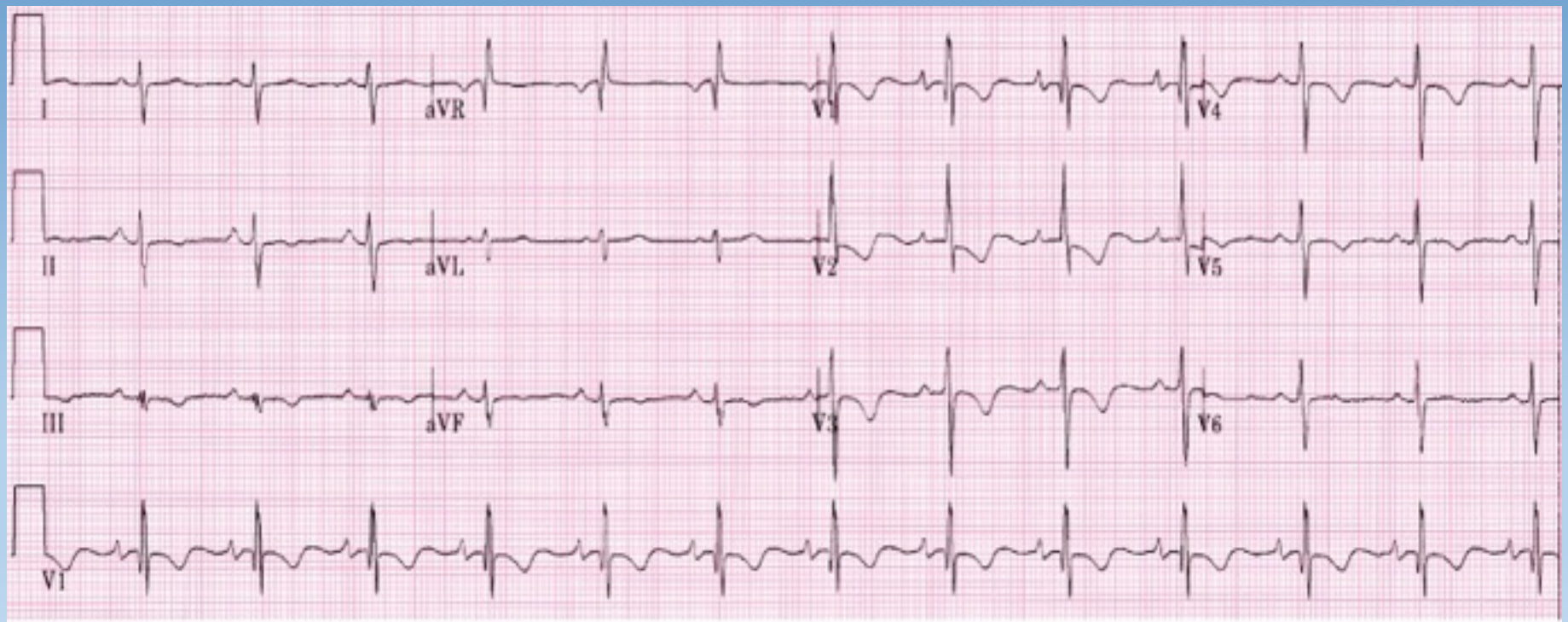


Right Ventricular Hypertrophy

- Right axis deviation
- qR in V1



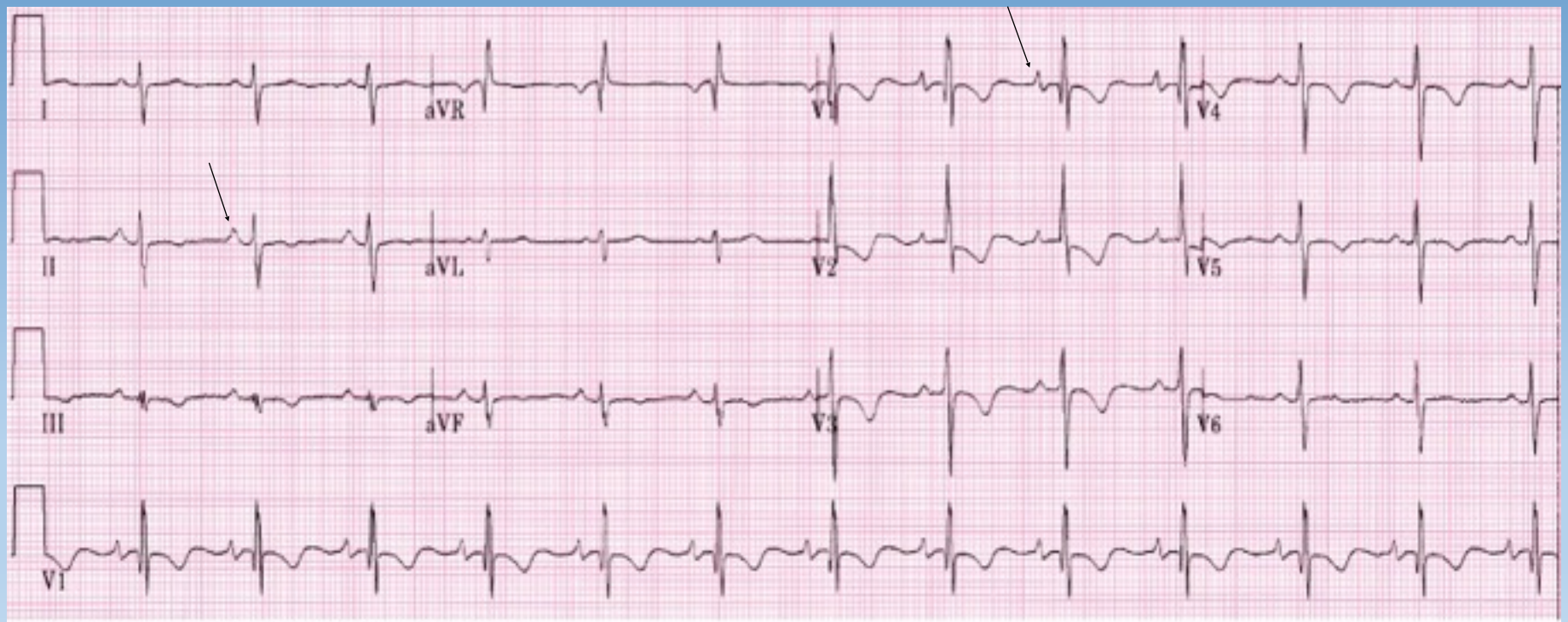
- Rate?
- Rhythm?
- Axis?
- Diagnosis?



Source: Knoop KJ, Stack LB, Storrow AB, Thurman RJ: *The Atlas of Emergency Medicine, 3rd Edition:* http://www.accessmedicine.com
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- RAD
- Dominant R wave
- Secondary ST-T changes in right precordial leads
- RAE

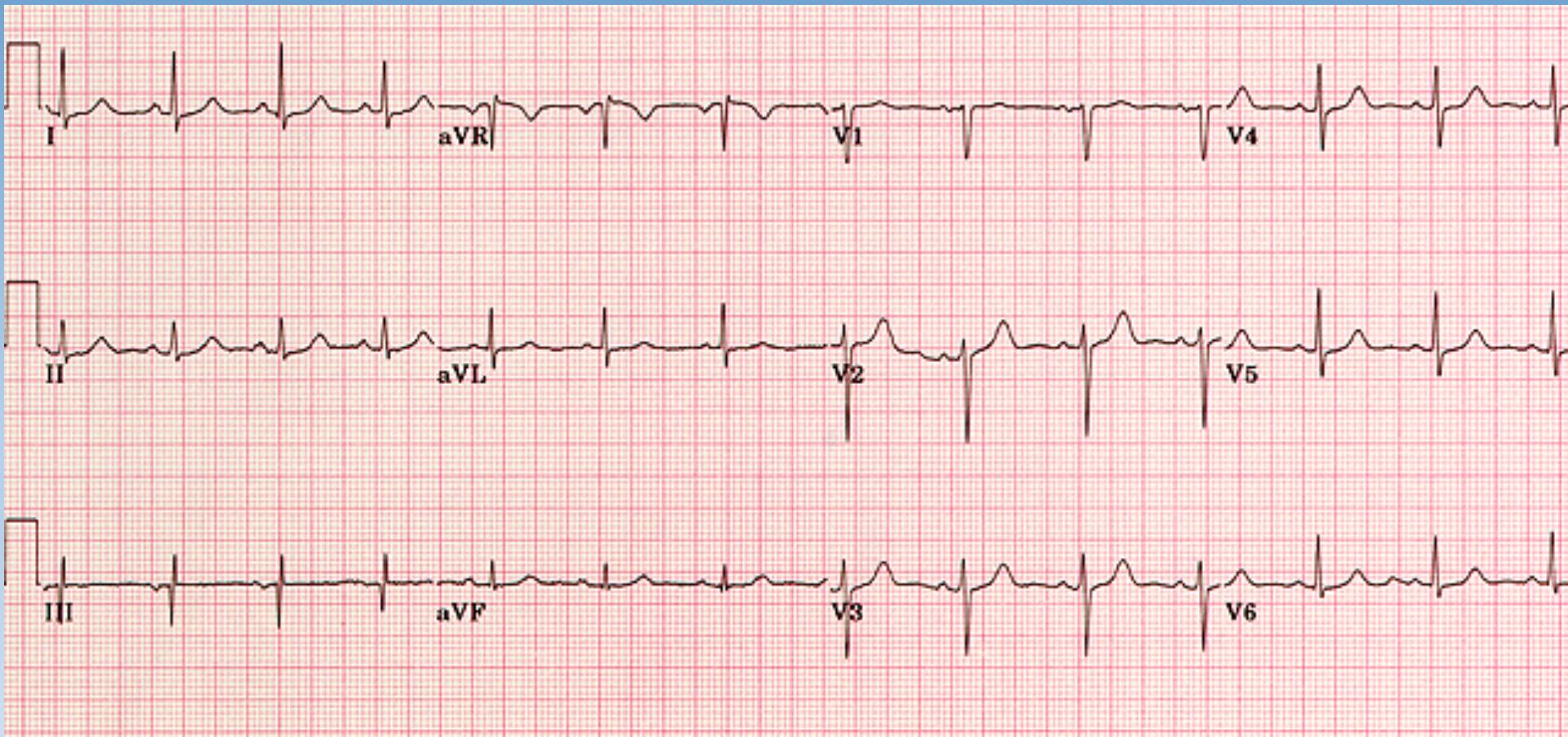
- Rate?78
- Rhythm? Normal sinus
- Axis? RAD
- Diagnosis? RVH



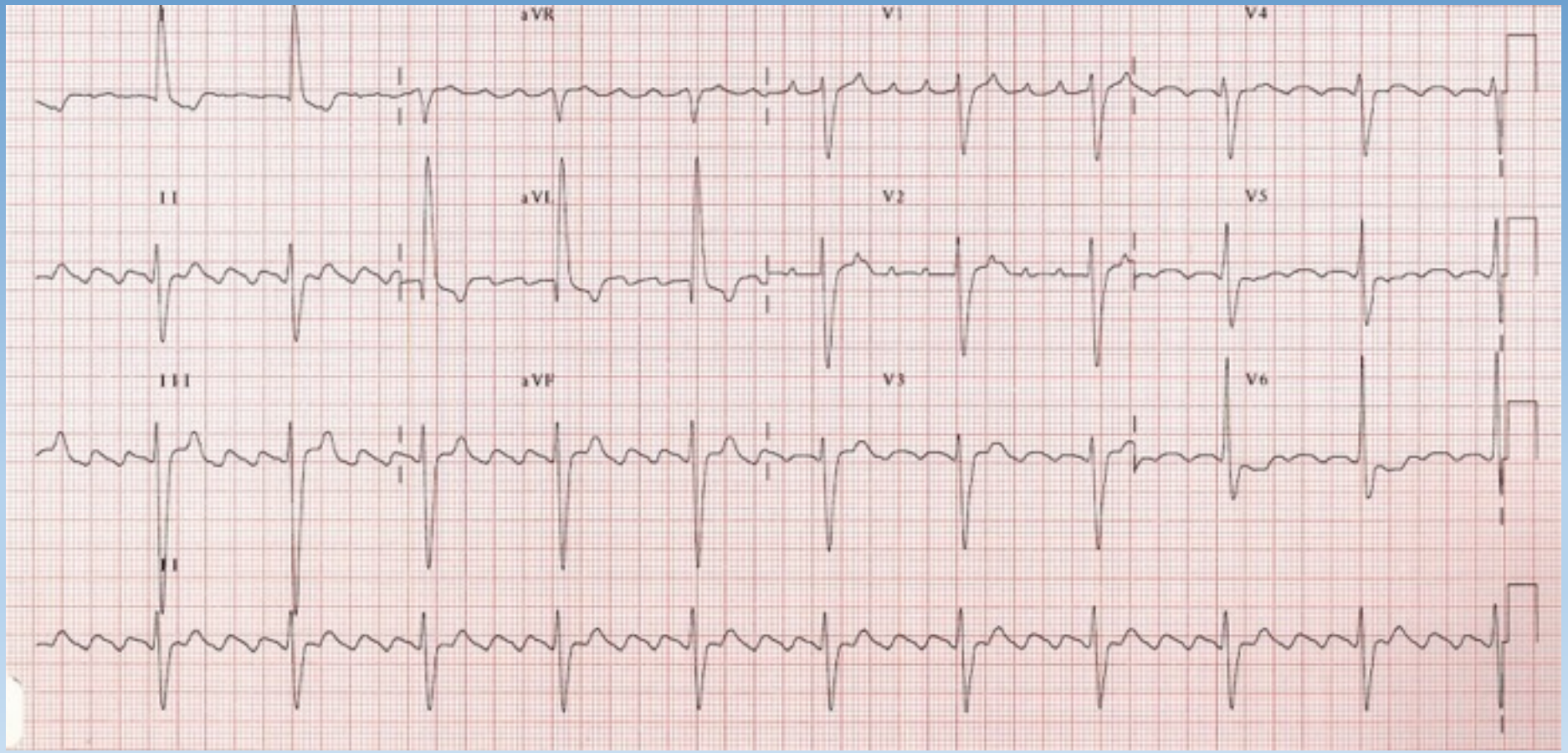
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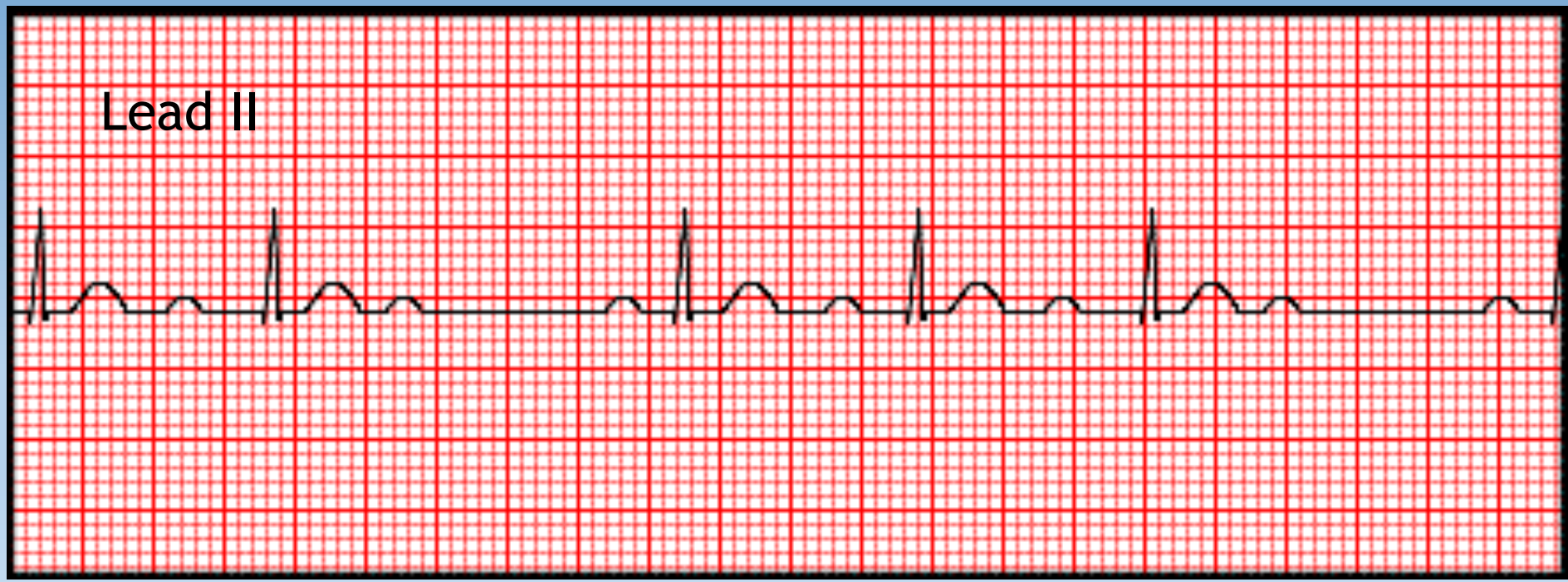
Test ECG #1



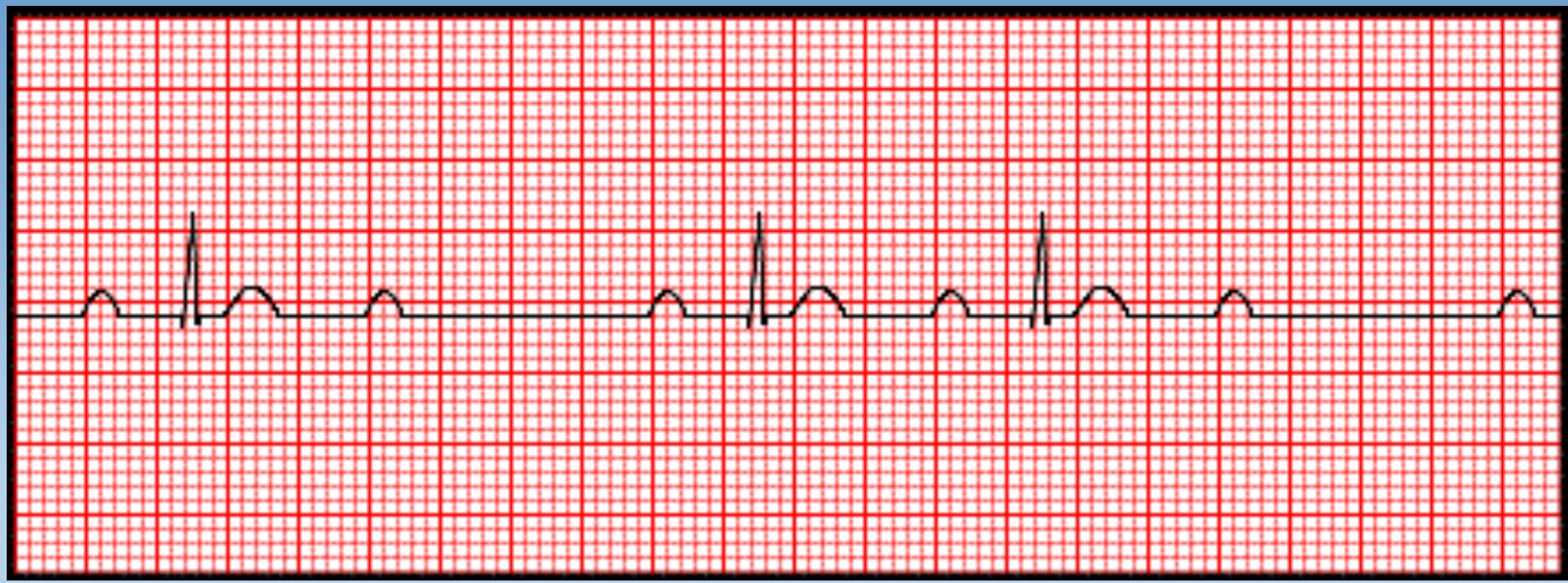
Test ECG #2



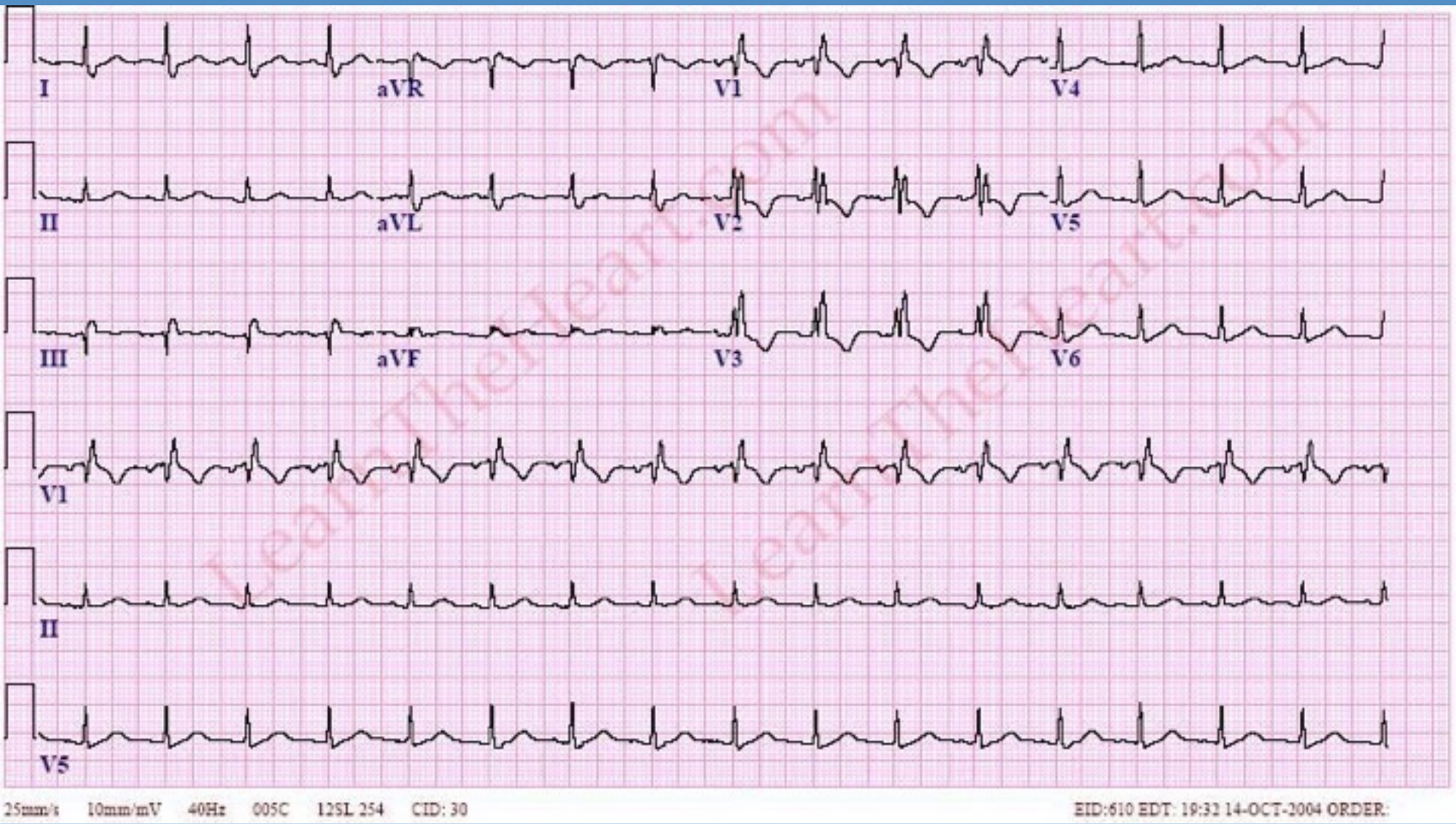
Test ECG #3



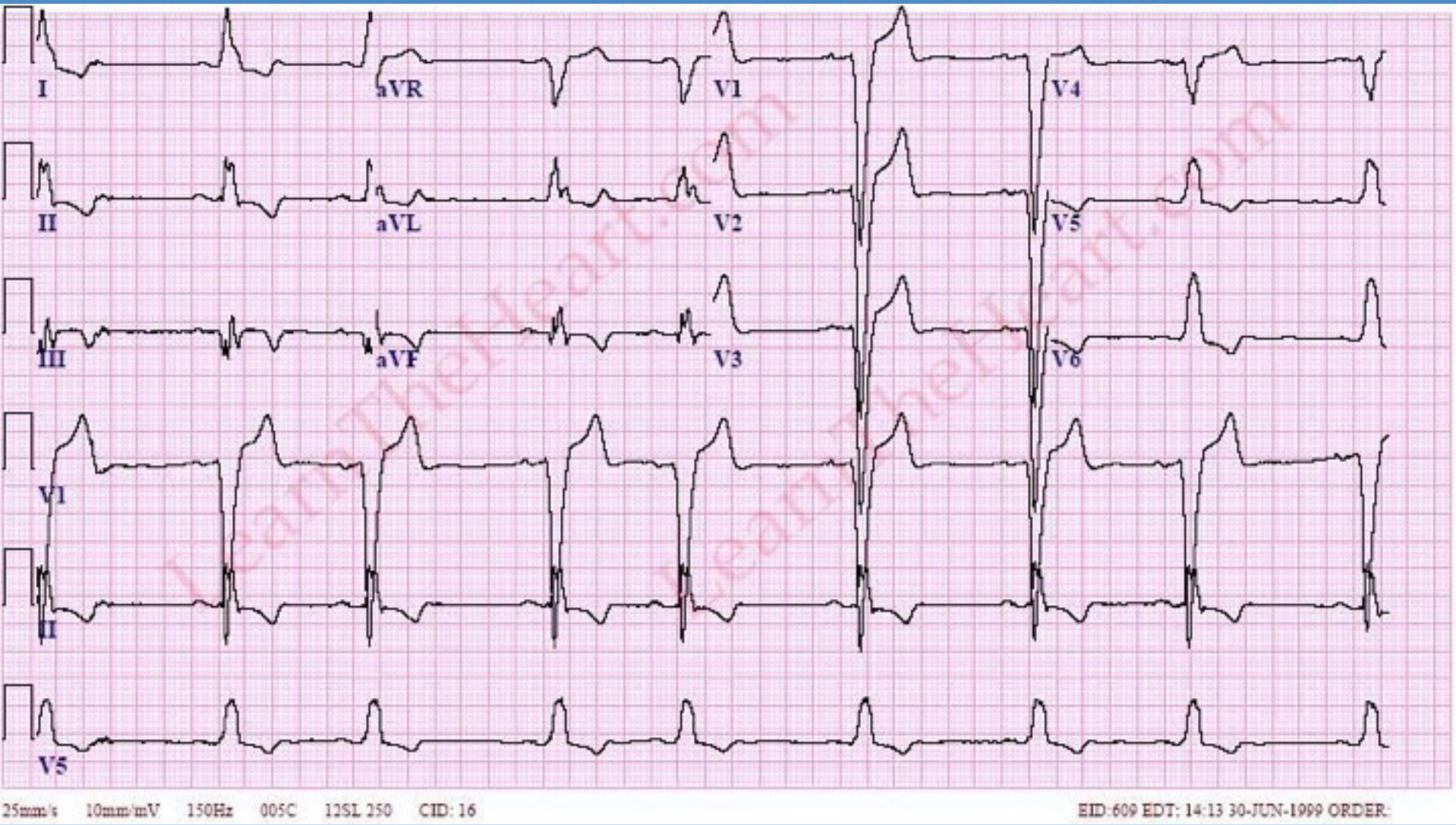
Test ECG #4



Test ECG #5


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
V1
II
V5
25mm/s 10mm/mV 40Hz 005C 12SL 254 CID: 30
EID:610 EDT: 19:32 14-OCT-2004 ORDER:

Test ECG # 6


I
aVR
V1
V4
II
aVL
V2
V5
III
aVF
V3
V6
V1
II
V5
25mm/s 10mm/mV 150Hz 005C 12SL 250 CID: 16
EID:609 EDT: 14:13 30-JUN-1999 ORDER:

Test ECG #7

